

CSCI 490: AI Alignment Research Engineering

Hunter College, City
University of New York

Department of Computer Science

CSCI 490: AI Alignment Research Engineering

Special Topics in Artificial Intelligence

Course Number	CSCI 490
Credits	3 credits
Semester	Fall 2026
Format	Online Synchronous
Meeting Day	[Day of week, e.g., Wednesday]
Meeting Time	[e.g., 6:00 PM - 9:00 PM ET]
Platform	Zoom (link distributed via Brightspace)
Instructor	[Instructor Name]
Email	[instructor@hunter.cuny.edu]
Office Hours	[Days, Times] via Zoom — link in Brightspace
Response Policy	

Emails answered within 48
hours on business days

This syllabus is subject to change. Any changes
will be announced via Brightspace and confirmed
in class.

Course Description

CSCI 490 is a project-centered course that builds practical and theoretical competencies in AI alignment research engineering. Drawing on the publicly available ARENA (Alignment Research Engineer Accelerator) curriculum, this course trains students to work on foundational problems in technical AI safety. Over twelve weeks, students progress from deep learning fundamentals through transformer mechanistic interpretability, reinforcement learning, large language model (LLM) evaluation, and alignment science.

Prerequisites

- Proficiency in Python (equivalent to CSCI 127 or higher)
 - Familiarity with linear algebra and basic calculus (MATH 155 / MATH 150 or equivalent)
 - Recommended: prior exposure to machine learning concepts (CSCI 381 or equivalent)
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Course Learning Outcomes

Upon successful completion of this course, students will be able to:

1. **Build and train** transformer language models end-to-end, including tokenization and autoregressive generation.
2. **Apply** mechanistic interpretability methods — including activation patching, circuit analysis, and sparse autoencoders — to analyze the internal computations of language models.
3. **Train** reinforcement learning agents using policy gradient and Q-learning methods, and apply RLHF to fine-tune a language model.

4. **Design and evaluate** multiple-choice LLM benchmarks and assess language model agent behavior.
 5. **Critically analyze** alignment-relevant phenomena such as emergent misalignment, reward hacking, and LLM psychology.
 6. **Communicate** technical AI safety research findings in written and oral formats appropriate for a research audience.
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Course Format — Online Synchronous

All sessions meet live via Zoom once per week during the scheduled meeting time. Attendance is expected for the full session. Each weekly session (~2 hours 30 minutes) is structured as follows:

Block	Duration	Activity
Lecture Part I	45 min	Instructor-led presentation and live demos
Break	15 min	—
Lecture Part II	45 min	Instructor-led presentation and live demos
Coding Lab	60 min	Students work on ARENA exercises; instructor circulates

Sessions will be recorded and posted to Brightspace within 24 hours for asynchronous review. Recordings do not substitute for live attendance.

Grading

Component	Weight
Weekly Coding Assignments (10 assignments, lowest 2 dropped)	35%
Quizzes (3 quizzes — Weeks 3, 6, 10)	15%
Midpoint Project (due end of Week 8)	20%

Component	Weight
Final Research Project & Presentation (due Week 12)	25%
Participation & Discussion	5%

Grading Scale

Grade	Range
A	93-100%
A–	90-92%
B+	87-89%
B	83-86%
B–	80-82%
C+	77-79%
C	70-76%
D	60-69%
F	Below 60%

Assessment Descriptions

Weekly Coding Assignments: Each week's ARENA exercises are submitted by 11:59 PM ET on the day before the next class session via Brightspace. Assignments are graded on correctness and code quality. You may use the ARENA AI assistant for hints, but submitted code must be your own.

Quizzes: Three short assessments (30-45 minutes) administered live via Brightspace at the start of class in Weeks 3, 6, and 10. Each covers material from the preceding unit. Closed-book; no external resources.

Midpoint Project: A written report (1,500-2,500 words) plus code demonstrating a mechanistic interpretability investigation of a language model circuit of your choosing. Due end of Week 8.

Final Research Project & Presentation: An individual or pair project (instructor approval required for pairs) exploring a topic from Chapters 2-4. Deliverables: GitHub repository with code, a 4-6 page technical report, and a 10-minute live Zoom presentation during the Week 12 session. Graded on technical depth, clarity, and originality.

Participation: Evaluated on engagement during live sessions, contributions to the class discussion board in Brightspace, and peer code review activities.

Course Policies

Attendance

Because this course meets only once per week, each session carries significant weight. Attendance at all sessions is strongly expected. More than **two unexcused absences** will result in a one-letter-grade deduction. If you anticipate a conflict, contact the instructor in advance. Documented emergencies (illness, family emergency) will be handled individually.

Late Work

Daily assignments submitted up to 24 hours late receive a maximum of 70%. Assignments will not be accepted more than 24 hours late except in cases of documented emergency. The midpoint and final projects have strict deadlines; late submissions require prior approval.

Academic Integrity

Hunter College regards acts of academic dishonesty (e.g., plagiarism, cheating, copyright infringement) as serious offenses against the values of intellectual honesty. The college is committed to dealing with such acts firmly and fairly. The penalty for an act of academic dishonesty may be a failing grade for the particular assignment and/or the course. Copies of the [CUNY Academic Integrity Policy](#) and the [Hunter College Honor Code](#) are available in the Provost's office.

Regarding AI tools: you may use AI coding assistants (GitHub Copilot, ChatGPT, the ARENA built-in assistant) as a learning aid to understand concepts and debug code. However, all submitted work must reflect your own understanding. Copying AI-generated solutions wholesale and submitting them without comprehension constitutes academic dishonesty.

Online Conduct

Students are expected to maintain professional behavior in Zoom sessions and Brightspace. This includes using your real name, keeping your camera on when feasible, and engaging respectfully. Hunter College has a zero-tolerance policy for bullying and cyberbullying.

College-Wide Statements

Accessibility

In compliance with the American Disability Act of 1990 (ADA) and with Section 504 of the Rehabilitation Act of 1973, Hunter College is committed to ensuring educational access and accommodations for all its registered students. Hunter College's students with disabilities and medical conditions are encouraged to register with the Office of AccessABILITY for assistance and accommodation. For information and appointment contact the Office of AccessABILITY located in Room E1214 or call (212) 772-4857 / TTY (212) 650-3230.

Title IX

Hunter College reaffirms its commitment to maintain an educational environment without sexual misconduct. Title IX of the Education Amendments of 1972 prohibits sex discrimination against any participant in an educational program or activity that receives federal funds. The act is intended to eliminate sex discrimination in education. Title IX covers discrimination based on sex or gender including sexual harassment, sexual assault, and sexual violence. If a student has a concern about such matters, they may call (212) 396-6080 or consult the [Hunter College Title IX page](#).

CUNY Policy on Sexual Misconduct

In compliance with the CUNY Policy on Sexual Misconduct, Hunter College reaffirms the prohibition of any sexual misconduct, which includes sexual violence, sexual harassment, and gender-based

harassment retaliation against students, employees, and visitors, as well as certain intimate partner violence.

Weekly Course Schedule

Key: One synchronous session per week (~3 hours). Total: 12 sessions across the semester. Assignments are due by 11:59 PM ET the day before the following class session.

Unit I — Fundamentals of Deep Learning (Weeks 1-3)

ARENA Chapter 0

Week 1 — PyTorch Essentials & Convolutional Networks

Topics: Course orientation; environment setup; PyTorch tensors, operations, and broadcasting; 1D and 2D convolution implemented from scratch; convolutional neural network (CNN) architecture.

ARENA Reference Ch0 §0.0–0.1

Assignment 1 Due Before Week 2 session

Outcomes: Students can manipulate tensors fluently and implement a convolution operation without library shortcuts.

Week 2 — ResNets & Transfer Learning

Topics: Residual network (ResNet) architecture and intuition; loading pretrained weights; fine-tuning on a downstream task; debugging training runs.

ARENA Reference Ch0 §0.2

Assignment 2 Due Before Week 3 session

Outcomes: Students can build a ResNet from scratch, load pretrained ImageNet weights, and fine-tune the model on a new dataset.

Week 3 — Backpropagation & Experiment Tracking | Quiz 1

Topics: Implementing automatic differentiation (autograd) from scratch; computational graphs; Weights & Biases (W&B) for experiment tracking and hyperparameter optimization.

ARENA Reference	Ch0 §0.3–0.4
Quiz 1	Administered at start of class (Unit I material)
Assignment 3 Due	Before Week 4 session

Outcomes: Students can write a backward pass by hand and instrument training runs with W&B dashboards.

Unit II — Transformer Architecture & Mechanistic Interpretability (Weeks 4–8)

ARENA Chapter 1

Week 4 — Transformer Architecture & Attention

Topics: The transformer as a mathematical object; self-attention and multi-head attention; positional encodings; layer normalization; residual stream framework.

ARENA Reference	Ch1 §1.1
Assignment 4 Due	Before Week 5 session

Outcomes: Students can describe every component of a transformer decoder and explain the flow of information through the residual stream.

Week 5 — Building & Training a Language Model

Topics: Implementing a GPT-style transformer decoder from scratch; tokenization; cross-entropy language modeling loss; autoregressive text generation with sampling strategies (greedy, top-k, nucleus).

ARENA Reference	Ch1 §1.2–1.3
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Assignment 5 Due Before Week 6 session

Outcomes: Students can train a small transformer LM end-to-end and generate coherent text from it.

Week 6 — TransformerLens & Induction Heads | Quiz 2

Topics: TransformerLens library for model inspection; activation hooks; identifying induction heads as an in-context learning circuit; head composition.

ARENA Reference	Ch1 §1.4-1.5
Quiz 2	Administered at start of class (Unit II Weeks 4-5 material)
Assignment 6 Due	Before Week 7 session

Outcomes: Students can use TransformerLens to extract and visualize attention patterns and identify induction heads programmatically.

Week 7 — Activation Patching & Circuit Discovery

Topics: Causal tracing and activation patching; path patching; indirect object identification (IOI) — replicating a full circuit analysis in GPT-2 Small; logit attribution.

ARENA Reference	Ch1 §1.6-1.7
Assignment 7 Due	Before Week 8 session
Midpoint Project Assigned	Topic proposal due end of this week

Outcomes: Students can run an activation patching experiment, attribute model behavior to specific attention heads, and interpret circuit diagrams.

Week 8 — Superposition, Sparse Autoencoders & Model Steering | Midpoint Project Due

Topics: Toy models of superposition and polysemanticity; sparse autoencoders (SAEs) — theory, training objectives, and feature visualization; steering model behavior via activation addition.

ARENA Reference	Ch1 §1.8–1.10
Assignment 8 Due	Before Week 9 session
Midpoint Project Due	11:59 PM ET end of this week

Outcomes: Students can train a sparse autoencoder, interpret learned features, and steer a language model toward a target behavior using activation vectors.

Unit III — Reinforcement Learning & RLHF (Weeks 9–10)

ARENA Chapter 2

Week 9 — RL Fundamentals, Bandits & Deep Q-Networks

Topics: Markov decision processes; policies, value functions, and Bellman equations; multi-armed bandits and exploration strategies (ϵ -greedy, UCB); Deep Q-Networks (DQN) with experience replay and target networks; OpenAI Gym environments.

ARENA Reference	Ch2 §2.1–2.3
Assignment 9 Due	Before Week 10 session

Outcomes: Students can implement and train a DQN agent on a standard Gym environment and articulate the exploration–exploitation tradeoff.

Week 10 — PPO & Reinforcement Learning from Human Feedback | Quiz 3

Topics: Policy gradient methods; Proximal Policy Optimization (PPO); reward model training; RLHF pipeline applied to the transformers built in Unit II.

ARENA Reference	Ch2 §2.4–2.5
Quiz 3	Administered at start of class (Unit III Week 9 material)
Assignment 10 Due	Before Week 11 session

Outcomes: Students can train a PPO agent and implement a full RLHF loop to fine-tune a language model toward a human preference signal.

Unit IV — LLM Evaluation & Alignment Science (Weeks 11-12)

ARENA Chapters 3 & 4

Week 11 — LLM Evaluation & Language Model Agents

Topics: Benchmark design principles; building a multiple-choice evaluation dataset from scratch; evaluating current models; constructing language model agent loops; tool use and agent evaluation methodologies.

ARENA Reference	Ch3 §3.1-3.2
Final Project Draft Due	End of this week (for peer review)

Outcomes: Students can design a rigorous LLM benchmark and build a basic LM agent with tool-use capability.

Week 12 — Alignment Science & Final Presentations

Topics: Emergent misalignment — sycophancy, reward hacking, and deceptive alignment; LLM psychology; the science of misalignment; modern AI safety research directions. **Final project presentations delivered live.**

ARENA Reference	Ch4 §4.1-4.3
Final Project & Report Due	11:59 PM ET day of final session

Outcomes: Students can articulate core alignment failure modes with concrete examples, situate the course material within the broader AI safety research landscape, and present an original research contribution to a technical audience.

Final Project Guidelines

The final project is an opportunity to explore one topic from the course in greater depth. Projects may replicate and extend a published result, apply a course technique to a new model or domain, or run an original experiment motivated by the course material.

Deliverables: 1. **GitHub repository** — well-documented code with a clear README 2. **Technical report** (4–6 pages, NeurIPS-style format) — introduction, related work, methods, results, discussion 3. **Presentation** (10 minutes + 5 min Q&A) — delivered live during the Week 12 session

Suggested project directions: - Replicate an IOI or greater-than circuit from the mechanistic interpretability literature in a different model - Train a sparse autoencoder and interpret the learned features - Implement and evaluate a novel RLHF reward model - Build a domain-specific LLM benchmark and evaluate 2–3 models - Run an experiment on emergent misalignment using model steering

Key Dates

Week	Event
Week 1	First class session; environment setup verified in class
Week 3	Quiz 1 (Unit I — Fundamentals)
Week 6	Quiz 2 (Unit II Weeks 4–5 — Transformer Architecture)
Week 7	Midpoint Project topic proposal due (end of week)
Week 8	Midpoint Project due at 11:59 PM ET
Week 10	Quiz 3 (Unit III Week 9 — RL Fundamentals)
Week 11	Final Project draft due (for peer review, end of week)
Week 12	

Week	Event
	Final Project Presentations (live, in-session) + Final Report due at 11:59 PM ET

Recommended Background Reading

These are not required but will deepen your understanding:

- Elhage et al. (2021). A Mathematical Framework for Transformer Circuits. Anthropic.
- Meng et al. (2022). Locating and Editing Factual Associations in GPT. NeurIPS.
- Christiano et al. (2017). Deep Reinforcement Learning from Human Preferences. NeurIPS.
- Hubinger et al. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. MIRI.
- Templeton et al. (2024). Scaling Monosemanticity. Anthropic.

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